

CMOS Two-Stage OTA Design and Analysis Using GPDK90nm

1. Objective

Design and analysis of a two-stage CMOS Operational Transconductance Amplifier (OTA) using GPDK90nm technology for high-gain analog applications.

2. Technology Details

Parameter Value

Technology GPDK90nm

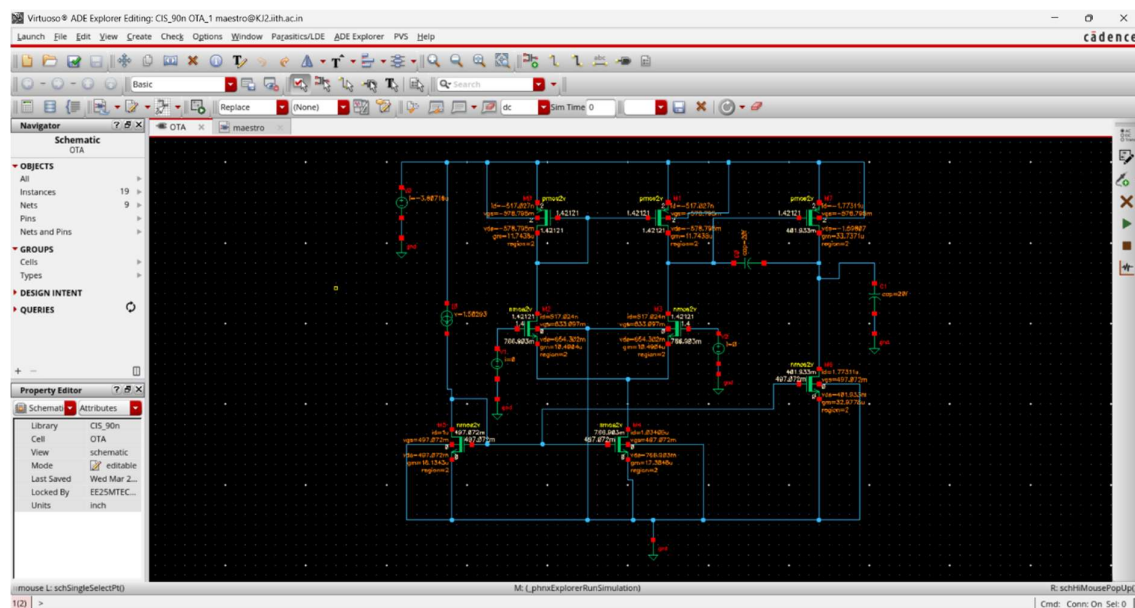
VDD 2 V

Process CMOS

Design Tool Cadence Virtuoso

Simulator Spectre

3. OTA Schematic



Circuit Description

- M2-M3 : Differential pair
- M0-M4-M5 : Current mirror bias
- M1-M7 : Active load

- M8 : Second gain stage
- Cc : Miller compensation capacitor
- CL : Load capacitor

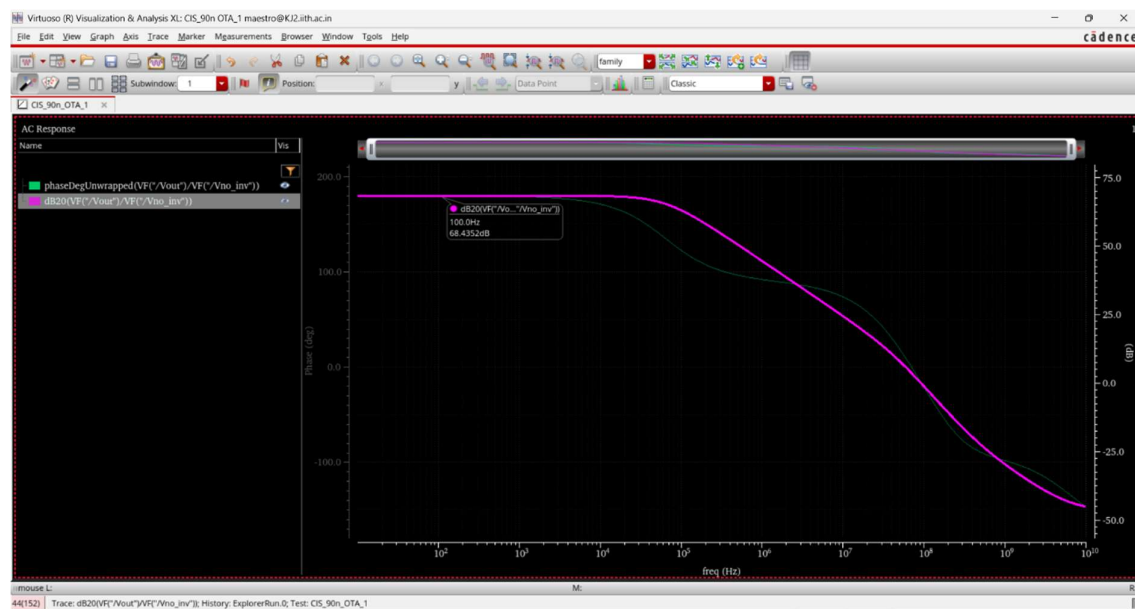
4. Design Parameters

Parameter	Value
Compensation Capacitor (Cc)	26 fF
Load Capacitor (CL)	20 fF
Supply Voltage	2 V
Bias Current	3 μ A

5. DC Operating Point Analysis:

- All MOSFETs operate in saturation(in region2)
- Proper DC biasing achieved.
- Output voltage lies within valid operating range.

6. AC response:



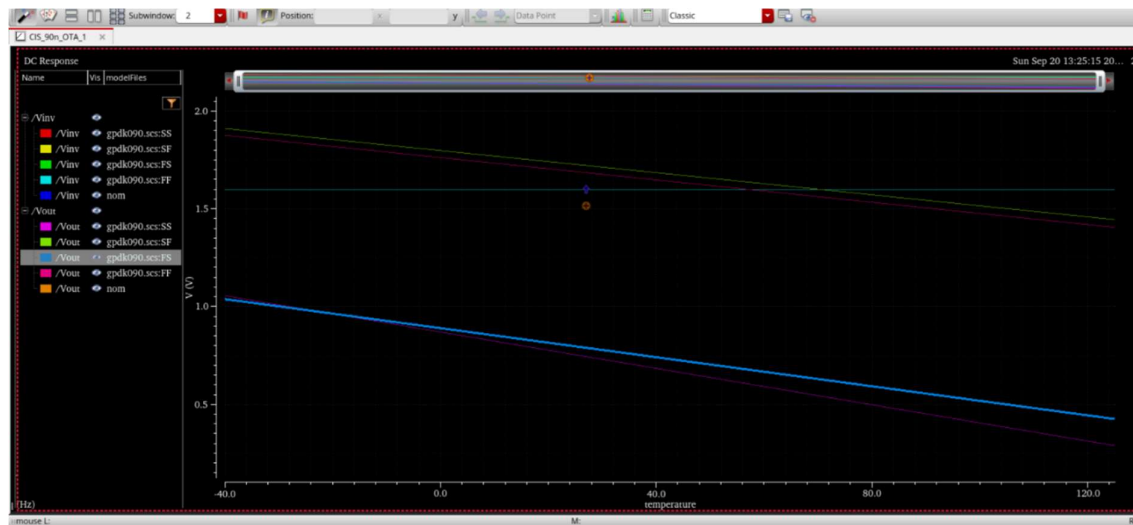
The gain observed is **68.42dB** which is decent enough to work this circuit as a suitable two stage amplifier.

7. PVT Analysis:

Process Corners: TT,SF,FS,FF,SS

Temperature Range : -40°C to 125°C

DC Response Across PVT



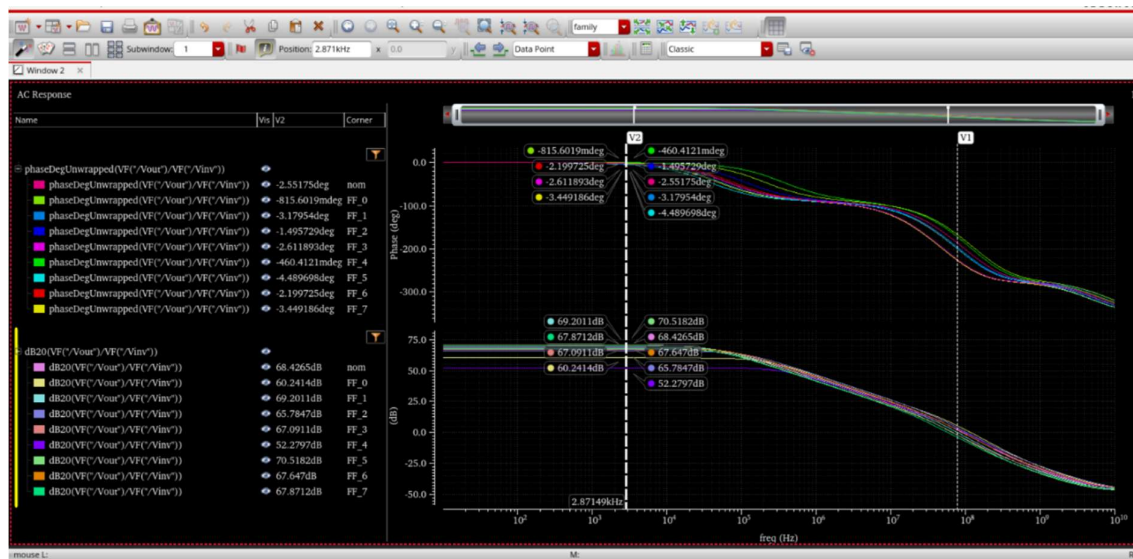
Observation:

- V_{inv} decreases with temperature.
- V_{out} decreases with temperature.
- Stable operating point maintained across all corners.

Result:

The OTA remains functional under all PVT conditions.

8. AC Analysis



AC Gain Results:

Temperature DC Gain

-40°C	52 dB
27°C	68.9 dB
125°C	70 dB

Gain Variation

$$\Delta Gain = 70 - 52 = 18dB$$

AC Response Observation

- High gain at low frequencies.
- Gain rolls off at higher frequencies due to dominant pole.
- Multiple poles observed from phase response.
- Gain maintained across all corners.

9. Conclusion

A two-stage CMOS OTA was successfully designed and simulated in GPD90nm technology using Cadence Virtuoso. The amplifier achieved a maximum gain of approximately 70 dB and maintained stable operation across process and temperature variations. DC and AC analyses verified the robustness of the design under worst-case conditions.